Experiment 8 - Develop a program to create reverse shell using TCP socket

Aim: To develop a Python program that demonstrates TCP socket communication between a client and a server.

Requirements:

- Hardware: Computer or laptop.
- Software: Python 3.x, any Python IDE (VS Code / IDLE), and a terminal.
- Module: Python `socket` module.

```
C:\MyFiles>python client.py
Connected to server
Enter message: Hi, this is Manikandan!
Server: Message received
Enter message: Have a Good day
Server: Message received
Enter message:
```

```
C:\MyFiles>python server.py
Waiting for connection...
Connected from: ('127.0.0.1', 53839)
Client: Hi, this is Manikandan!
Client: Have a Good day
```

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```python
import socket

HOST = "0.0.0.0"
PORT = 5000

server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind((HOST, PORT))
server.listen(1)

print("Waiting for connection...")

conn, addr = server.accept()
print("Connected from:", addr)

while True:
    message = conn.recv(1024).decode()

    if not message:
        break

    print("Client:", message)

    reply = "Message received"
    conn.send(reply.encode())

conn.close()
server.close()
```

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```python
import socket

SERVER_IP = "127.0.0.1"
PORT = 5000

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect((SERVER_IP, PORT))

print("Connected to server")

while True:
    message = input("Enter message: ")

    if message.lower() == "exit":
        break

    client.send(message.encode())

    response = client.recv(1024).decode()
    print("Server:", response)

client.close()
```